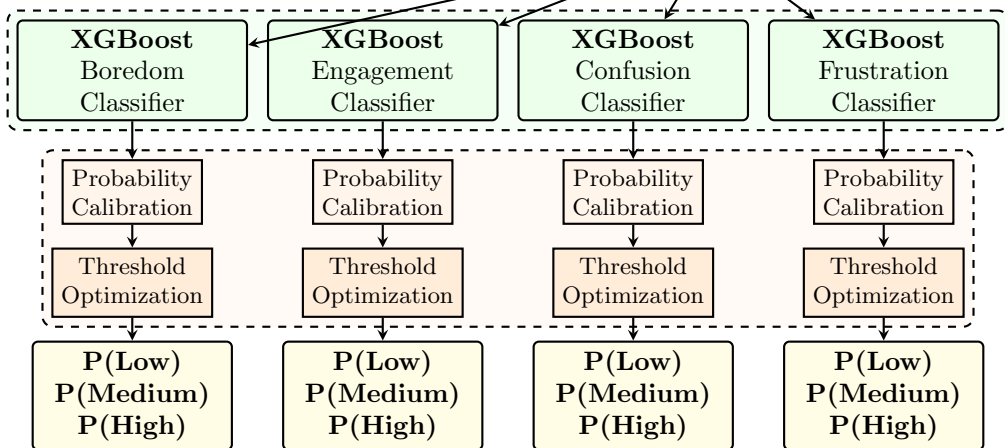
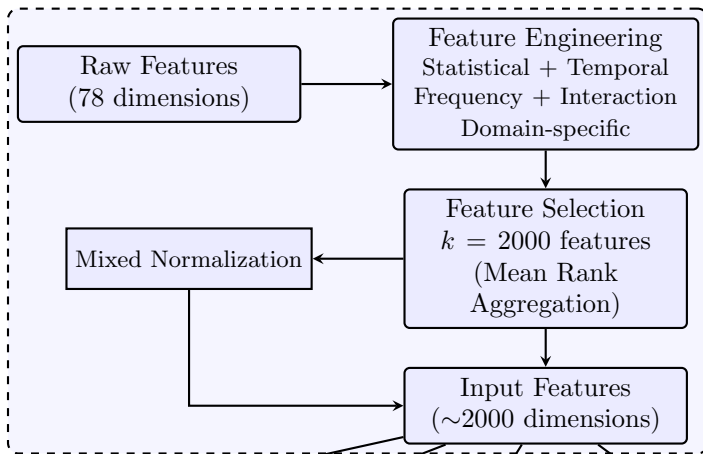




#### XGBoost Hyperparameters:

- max\_depth=8
- learning\_rate=0.05
- n\_estimators=500
- subsample=0.8
- colsample\_bytree=0.8
- min\_child\_weight=3
- gamma=0.1
- reg\_alpha=0.1 (L1)
- reg\_lambda=1.0 (L2)
- objective=multi:softprob
- early\_stopping\_rounds=50

#### Class Imbalance Handling:

- SMOTE Oversampling:  $k=5$  neighbors
- Class Weights: Inverse frequency
- Focal Loss:  $\alpha=0.25, \gamma=2.0$

#### Calibration & Thresholding:

- Calibration Method: Isotonic regression
- Threshold Metric: F1-macro
- Search Range: [0.2, 0.8]